

Performance Work Statement (PWS)

Title: Computer-Based Testing for the Cryptographic Validation Program (CVP) 2025-27

I. Background

The National Institute of Standards and Technology (NIST) promotes the U.S. economy and public welfare by providing technical leadership for the nation's measurement and standards infrastructure. The Cryptographic Module Validation Program (CMVP) was developed to support the federal user communities for strong, independently tested, and commercially available cryptographic modules. Through this program, the CMVP works with international government, public and private sectors as a part of the cryptographic community to achieve standards-based security and assurance of correct implementation. Federal agencies are required to use validated cryptographic modules for the protection of sensitive unclassified information. The goal is to provide those agencies, as well as other users, with a security metric list to use in procuring and deploying validated cryptographic modules. In other words, all non-DOD Federal Agencies must use software that has been validated by the CMVP when it comes to security related to cryptographic modules.

In advancing its mission objectives, NIST must validate cryptographic modules and cryptographic algorithms, and ensure independent, National Voluntary Laboratory Accreditation Program (NVLAP) accredited laboratories meet competence and performance criteria for conducting algorithm and module testing. Cryptographic and Security Testing (CST) Laboratories are independent laboratories accredited by NVLAP. CST Labs verify each module meets a set of testable cryptographic and security requirements, with each CST laboratory submission reviewed and validated by CMVP. These labs are located around the world, including North America, Asia, Australia, and Europe.

To maintain quality in the CMVP an exam is given to all testers. They must pass this exam to evaluate and test the modules. Currently the exam is 100 questions which the CMVP has developed over many years. The exam cannot be copied or removed from the test site so the questions will not become public knowledge. In addition, strict ID verification must be conducted to ensure the test takers are who they say they are.

II. Purpose

The purpose of this requirement is to publish, administer, and update computer-based competency tests to testing centers near NVLAP accredited laboratories.

Objectives:

NIST's Computer Security Division's (CSD) objective for this tasking includes administering computer-based competency exams to NVLAP accredited laboratory testers for the Security Testing, Validation and Measurement (STVM) Group's cryptographic validation programs.

- A secure location for taking the exam
- ID verification capabilities
- On-line registration capabilities

- Administering computer-based competency exams to NVLAP accredited laboratory testers for the Security Testing, Validation and Measurement (STVM) Group's cryptographic validation programs
- Test centers to be located within a reasonable distance from the labs
- The capability to add new testing locations as new CST Labs come online

Period of Performance

The period of performance shall be a base period of twelve (12) months plus one (1) 12-month option periods as follows:

Base Period: May 1, 2025 through April 30, 2026
 Option Period 1: May 1, 2026 through April, 30, 2027

Place of Performance

The place of performance shall be at the contractor's facility.

III. Specific Requirements

Tasks

The contractor shall provide all labor, project oversight, administration and technical execution of the tasks and deliverables identified in this Performance Work Statement (PWS). The contractor shall be responsible for maintaining accurate records of project activities and shall provide the support services described below.

1.1 Test Publishing and Administration (Base and Option Period)

The contractor shall:

1. Prepare/update candidate instructions that are appropriate for NIST to distribute to the NVLAP accredited laboratories. The instructions shall be specific to the NIST test. For example, the information may include the following:
 - a. Test duration
 - b. Materials that may be brought to testing facility
 - c. Versions of documents that will be available
 - d. How to register
2. Publish and administer the FIPS 140-3 competency exam via a computer-based system. This shall include the following:
 - a. Provide supporting documents to test candidates during administration of the exam. On a periodic schedule, NIST will supply contractor with documents that are authorized for use during the exam
 - b. Contractor shall also provide a means for candidates to address issues/questions during the test

3. Provide evidence of secure testing centers worldwide, of which 85% shall be within 50 miles of NVLAP accredited laboratories currently located in the US, Canada, Japan, Germany, Malaysia, Spain, France, and Taiwan. A current list of NVLAP accredited labs may be found in [Appendix A](#) at the end of this document. More CST labs may be added during this contract period which will increase the number of testing locations needed. NIST will notify the contractor in order to request a new test center
4. Schedule candidate testing
5. Contractor shall collect a \$400 fee from candidates. It is anticipated that 30 to 40 candidates will require taking the exam per year
6. Verify candidates with NIST to ensure candidates are eligible to sit for the test
7. Verify/authenticate candidate's identity
8. Protect exam content from unauthorized removal by monitoring candidates during the entirety of the exam
9. Secure candidates test results and share those results only with NIST
10. NIST will notify candidate of results

1.2 Exam Maintenance (Option Period Only)

The contractor shall:

1. Work collaboratively with NIST to review the FIPS 140-3 competency exam and update up to 10% of the test questions, of each exam, once a year. Updated raw questions will be provided to the contractor by NIST, but will require:
 - a. Vetting new questions for bias and difficulty and updating scoring appropriately
 - b. Removing and/or editing questions
 - c. Balancing and assessing questions
 - d. Updating tests to incorporate changes

1.3 Project Management & Reporting

1.3.1 Kick-off Meeting (Base Period Only)

The contractor shall attend a virtual kick-off meeting within 10 business days of award where the TO PM will identify Key Personnel and ask the government any questions they may have at that time.

1.3.2 Project Reporting (Base and Option Period)

The Contractor shall provide a monthly report, via email, of the status of the project to the government Technical Point of Contact (TPOC) and Contracting Officer's Representative (COR), and other stakeholders identified by NIST. The monthly status reports shall detail

activities accomplished, deliverables completed, outstanding deliverables, any delays, reasons for delays and proposed resolutions, and recommendations.

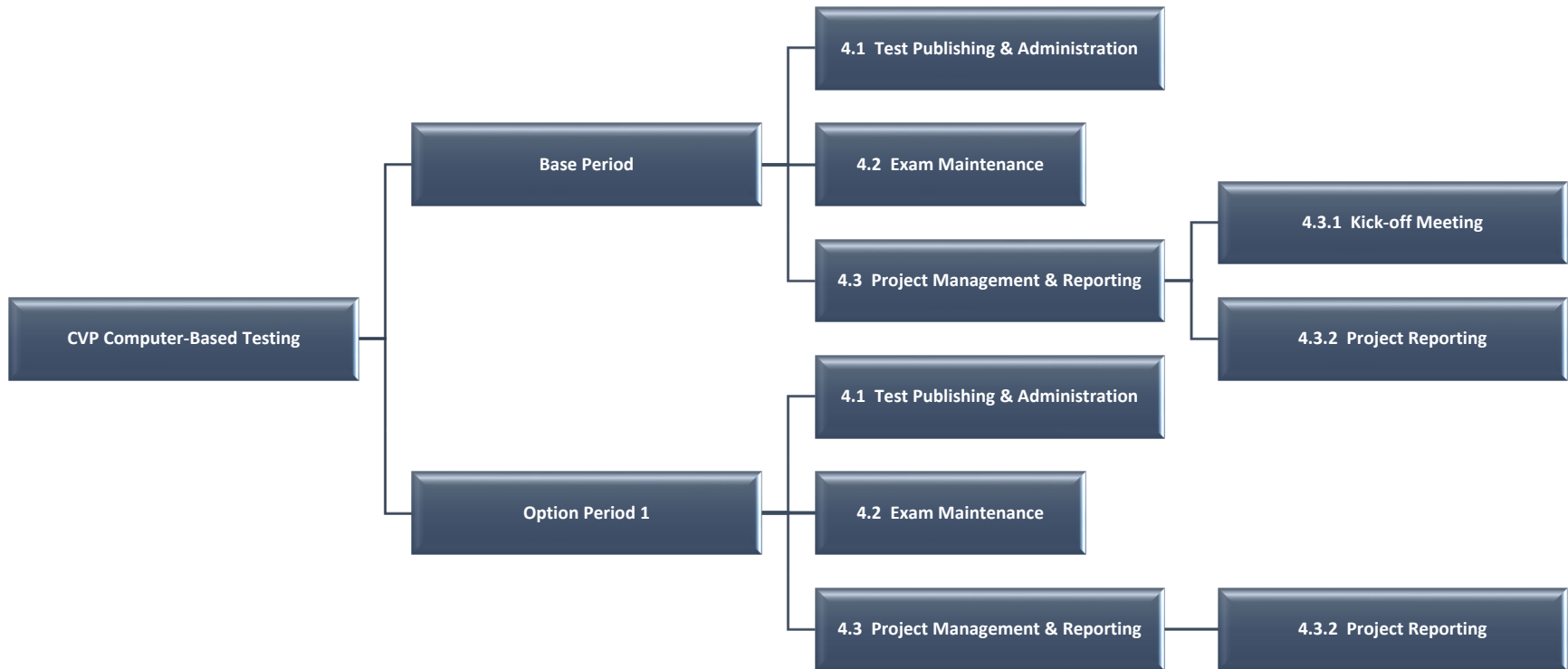


Figure 1: Government Work Break Down Structure (WBS)

IV. General Requirements

Government-Furnished Property, Material, Equipment, or Information (GFP, GFM, GFE, or GFI)

No Government-furnished property or equipment will be required as a part of this effort.

The government will furnish information as needed to develop and update exams. The Government will provide electronic copies of any relevant documents necessary to complete the tasks.

Key Personnel

It is anticipated that one (1) Contractor Key Personnel be required and shall meet the following minimum qualifications.

- Computer-Based Test Analyst (or equivalent) and shall have 2 years of experience in developing computer-based test questions from questions supplied by the customer.

V. Schedule of Deliverables and Performance Requirements Summary (PRS)

All deliverables shall be delivered to NIST via secure methods as directed by the Technical Lead or the COR. The COR will evaluate the deliverables for completeness and will either accept or reject within 10 days of contractor submission. All deliverables shall be provided to the COR.

		Description	Projected Completion Date	Media / Desired Output / Performance Standard	Monitoring Method
Base and Option Period 1					
D1	4.1.1	Instructions for candidates prepared/updated	Delivery date within 2 months of award	• MS Outlook / MS Word / Contractor format • Candidate instructions that NIST will distribute to the NVLAP laboratories that are specific to the NIST test • Instructions shall be free of grammatical and typographical errors	TPOC and COR Review

		Description	Projected Completion Date	Media / Desired Output / Performance Standard	Monitoring Method
Base and Option Period 1					
				- Information may include, but not be limited to, the following: ➤ Test duration ➤ Materials that may be brought to testing facility ➤ Versions of documents that will be available ➤ How to register	
D2	4.1.2	FIPS 140-3 Competency exam published & administered via a computer-based system	Test administration to begin immediately after Base award or award of modification	- Contractor format - Competency exam shall consist of two 100 question exams, of which one will be chosen at the time of testing to be administered via a computer-based system - Publish and administer exam as developed in collaboration with the NIST technical team - Provide NIST supplied supporting document to test candidates during administration of the exam ➤ Provide a means for candidates to address issues/questions during the exam	TPOC and COR Review
D3	4.1.3	Evidence of security measures taken at testing centers	To begin immediately after Base award or award of modification	- MS Outlook / MS Word / Contractor format during monthly status reports at a minimum - Evidence of a secure testing environment - Emails or reports documenting security measures that are taken at each testing facility to prevent unapproved persons from taking the test as well as to prevent the theft of text questions/materials	TPOC and COR Review
D4	4.1.4	Evidence of candidate test scheduling	To begin immediately after Base award or award of modification	- MS Outlook / MS Word / Contractor format during monthly status reports at a minimum - Monthly report of upcoming scheduled tests emailed to COR & TPOC on first of each month and available within 2 business days upon request if there are questions/problems with a candidate getting scheduled	TPOC and COR Review

		Description	Projected Completion Date	Media / Desired Output / Performance Standard	Monitoring Method
Base and Option Period 1					
D5	4.1.6	Candidate test eligibility verified	To begin immediately after Base award or award of modification	- MS Outlook / MS Word - Notification from contractor, via report or email, that a candidate wishes to schedule a test - TPOC verifies candidate's eligibility and informs contractor whether candidate is eligible to take the test	TPOC and COR Review
D6	4.1.7	Candidate identity verified/ authenticated	To begin immediately after Base award or award of modification	- MS Outlook / MS Word / Contractor format - Report of measures taken to authenticate people before the test is administered so that only verified candidates are tested - Records for each verified candidates shall be kept on file should case problems arise	TPOC and COR Review
D7	4.1.8	Exam content protected by monitoring candidates for entirety of exam	To begin immediately after Base award or award of modification	- MS Outlook / MS Word / Contractor format - Secure exam content so that it is not replicated outside the testing environment by laboratories - A written record per applicant verifying that the applicant was monitored for the entirety of the test and there was no observation of the candidate copying or otherwise duplicating the exam material	TPOC and COR Review
D8	4.1.9	Candidate test results secured and shared only with NIST	To begin immediately after Base award or award of modification	- MS Outlook / MS Word / Contractor format - Tests are calculated and results are sent to NIST for each candidate - A written record of statistics from the exam for a candidate - Contractor shall not share results with the candidate, they shall share the results with only NIST and NIST shall inform the lab/candidate of their score and results	TPOC and COR Review
D9	4.2	FIPS 140-3 competency exam reviewed & updated up to 10% each exam	Once a year for each Option Period	- Contractor's format - Update of exam shall include: ➤ Review test and update up to 10% of test questions,	TPOC and COR Review

		Description	Projected Completion Date	Media / Desired Output / Performance Standard	Monitoring Method
Base and Option Period 1					
				twice a year ➤ Updated raw questions will be provided by NIST but will require: ▪ Vetting new questions for bias and difficulty and updating scoring appropriately ▪ Removing and/or editing questions ▪ Balancing and assessing the difficulty of each question in order to assist NIST with evaluating questions ▪ Updating test to incorporate changes ▪ Assembling a balanced test from a pool of questions or multiple tests created that are managed by the vendor in order to prevent candidates and/or laboratories from memorizing ▪ Determination of passing score ▪ Working with NIST to assess post test score patterns and/or issues ▪ Editing to be free of grammatical and typographical errors • Initial delivery within 6 months of award, then twice yearly	
D10	4.3.1	Kick-off Meeting (Base Period only)	Within 10 business days after award	• MS Word / MS Excel / MS PowerPoint • The Kick-off meeting shall be no later than 10 business days after Base Period award • Electronic copy of the agenda shall be delivered to the COR at least 2 business days before the meeting • The kick-off meeting shall be utilized to introduce all members of the contractor's team, review all requirements, deliverables, and schedule.	

		Description	Projected Completion Date	Media / Desired Output / Performance Standard	Monitoring Method
Base and Option Period 1					
D1 1	4.3.2	Project Reporting	Monthly	• MS Word / MS Excel / MS PowerPoint submitted via email • Monthly updates shall be submitted to the COR before each monthly meeting • Reports shall be legible and of a professional quality • Any problems with the work, current or anticipated, shall be clearly reported	TPOC and COR Review

Appendix A

Current list of Laboratories

Current list of National Voluntary Laboratory Accreditation Program (NVLAP) laboratories obtained from the CSRC website: <https://www-s.nist.gov/niws/index.cfm?event=directory.search#no-back>

Lab Name	City	State	Country
Acumen Security	Rockville	MD	US
ADVANCED DATA SECURITY	San Jose	CA	US
AEGISOLVE, Inc.	Mountain View	CA	US
Apple Inc. SECLAB	Cupertino	CA	US
Asia Pacific IT Laboratory, TUV NORD	Kaohsiung		TW
ATSEC information security corporation	Austin	TX	US
Booz Allen Hamilton Cyber Assurance Testing Laboratory	Laurel	MD	US
Cisco Systems Automated Cryptographic Validation Protocol Lab	Morrisville	NC	US
CyberSecurity Malaysia Cryptographic Evaluation Laboratory	Selangor		MY
DEKRA Cybersecurity Certification Laboratory	Sterling	VA	US
Dekra Testing and Certification S.A.U.	San Sebastian de los Reyes, Madrid		ES
ECSEC Laboratory Inc.	Tokyo		JP
EWA – Canada	Ottawa, ON		CA
Google LLC	Mountain View	CA	US
Gossamer Security Solutions	Columbia	MD	US
Intel CST Lab	Hillsboro	OR	US
IT Security Center	Chiyoda-ku, Tokyo		JP

Appendix A
Current list of Laboratories

Leidos Accredited Testing & Evaluation (AT&E) Lab	Columbia	MD	US
Lightship Security Inc.	Ottawa, Ontario		CA
NXP ACVP Laboratory	Toulouse		FR
Penumbra Security, Inc.	Clackamas	OR	US
SERMA SAFETY AND SECURITY	Pessac Cedex		FR
Teron Labs	Deakin		AU
TUVIT Evaluation Body for IT Security	Essen		DE
UL Verification Services, Inc.	San Luis Obispo	CA	US